

Editorial

HIV: The Math

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Roughly 1% of the world's adult population is infected with HIV, which currently results in 2.8 million deaths per year (3% of all deaths and almost three times as many as malaria).

What can mathematics offer in the struggle against this disease? Mathematics itself offers no protection and no cures. However, like claims early on that "everyone" can get AIDS, and revelations of false assurances of safety of the blood supply in the U.S. and France, mathematics can definitely help add to the scare effect. Many years ago, I rejected for publication in this *Journal* a paper by a student team that projected that most American adults would be infected by HIV—or dead from it—by now. Both the modeling and the conclusions were unsound. The dynamic of HIV, however, has since taught us again an old lesson, that fear is weaker than desire. (This *Journal* did publish a UMAP Module on HIV recently [Isihara 2005], dealing mainly with immunological aspects.)

What mathematics can do is project immediate past and current trends, to reveal what the future could be without basic change. In the case of HIV, the results are encouraging for some countries and discouraging for others. The teams in this year's Interdisciplinary Contest in Modeling were asked to focus their modeling on "critical" countries, with the teams determining for themselves the meaning of "critical" and selecting the countries. The Outstanding papers published here unsurprisingly focus on many of the same countries (ones with large HIV-positive populations or with a large proportion of their population HIV-positive) and project, in human and in economic terms, the results of different strategies of intervention.

Mathematics can certainly help quantify the varied consequences of differing policies, such as for preventing HIV. But many people mistrust the predictions of mathematics because they mistrust mathematics. And they mistrust

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mathematics because they never understood it nor learned to appreciate its relevance.

The thousands of interested students who take part in the ICM and MCM are one prong of COMAP's efforts to promote applications in mathematics instruction. However, it is those millions of students mistrustful of mathematics whom COMAP tries hardest to reach, with its high school mathematics textbooks [COMAP 2000; 2002; 2007; Crisler and Froelich 2006], its college-level textbook [Garfunkel et al. 2006], and other initiatives.

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About the Editor



Paul Campbell graduated summa cum laude from the University of Dayton and received an M.S. in algebra and a Ph.D. in mathematical logic from Cornell University. He has been at Beloit College since 1977, where he served as Director of Academic Computing from 1987 to 1990. He is Reviews Editor for *Mathematics Magazine* and has been editor of *The UMAP Journal* since 1984. He is a co-author of the COMAP-sponsored book of applications-oriented collegiate mathematics *For All Practical Purposes* (7th ed. W.H. Freeman 2006), already used by more than half a million students.

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